MATHEMATICS

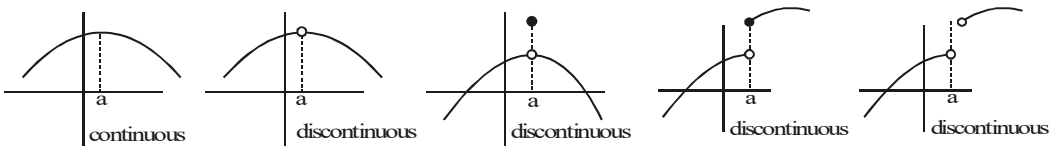
CONTINUITY & DIFFERENTIABILITY THEORY SHEET ENGLISH

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CONTINUITY

1.1 DEFINITION OF CONTINUITY

Graphical definition : While sketching the graph if we do not need to lift our hand then we can say that function is continuous at that point.



Mathematical Definition :

A function f(x) is said to be continuous at x = a, if $\lim_{x\to a} f(x) = f(a)$.

Symbolically f is continuous at x = a if $\lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(a+h) = f(a), h>0$

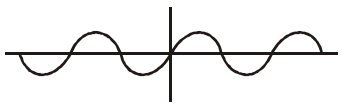
$LHL_{x=a} = RHL_{x=a} = f(a)$

Q. Discuss continuity of following functions with the help of graphs

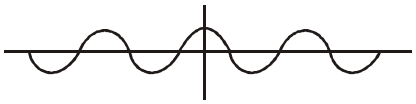
1. $y = \sin x$ 2. $y = \cos x$ 3. $y = [x]$ 4. $y = \{x\}$

5. $y = |x|$ 6. $y = \text{sgn}(x)$

Sol. (1) $y = \sin x$

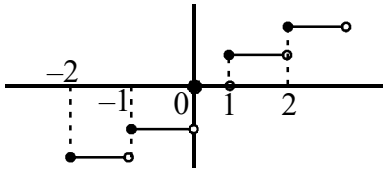


always continuous

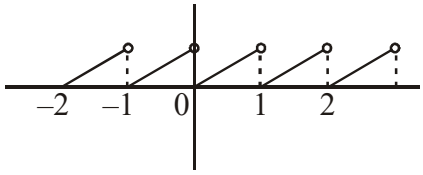
(2) 

always continuous

(3) $y = [x]$



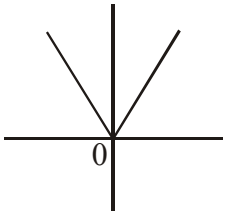
(4) $y = \{x\}$



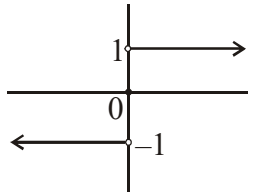
DC at $x \in I$

C at $x \notin I$

(5) $y = |x|$



(6)



DC at $x = 0$

C at $x \neq 0$

1.2 IMPORTANT POINTS RELATED TO CONTINUITY

1. If function is continuous at $x = a$, then $x = a$ must be in domain of function.
2. Continuity implies existence of limit but converse need not to be true.
3. Function f(x) is said to be left continuous at x= a if $\lim_{x \to a^-} f(x) = f(a)$
4. Function f(x) is said to be right continuous at x = a if $\lim_{x \to a^+} f(x) = f(a)$
5. Continuity is always discussed in domain.

   Eg. : $f(x) = \frac{1}{x}$

   (i) Always continuous : True (ii) Discontinuous at $x = 0$: True

   (iii) Always continuous in $[-1, 1]$: False
6. All Polynomials, Trigonometrical functions, exponential and Logarithmic functions are continuous in their domain .
7. If f(x) and g(x) both are continuous at $x = a$, then function $(f + g)$, $(f - g)$, $(f \times g)$ and $\left(\frac{f}{g}\right)$ will be continuous at x= a.
8. If f(x) is continuous at $x = a$ and g(x) is discontinuous at $x = a$ then,

   (i) function $(f + g)$ and $(f - g)$ will be discontinuous at $x = a$.

   (ii) Nothing definite can be said about continuity of $(f \times g)$ and $\left(\frac{f}{g}\right)$ at $x = a$.

E.g. $f(x) = x[x]$ Check at (i) $x = 0$ (ii) $x = 1$

9. If both functions f(x) and g(x) are discontinuous at $x = a$, then nothing definite can be said about continuity of $(f+g)$, $(f-g)$, $(f \times g)$ and $\left(\frac{f}{g}\right)$ at $x = a$.

E.g. (i) $h(x) = [x] + \{x\}$ at $x = 0$, at $x = 1$

(ii) $h(x) = [x] . \{x\}$ at $x = 0$, at $x = 1$

Check continuity of following functions (Q1 to Q3)

1 $f(x) = \begin{cases} \sin\frac{\pi x}{2}, & x < 1 \\ [x] & x \geq 1 \end{cases}$ at $x = 1$, where [ ] denotes greatest integer function.

Ans continuous

Sol. : $f(x) = \begin{cases} \sin\frac{\pi x}{2}, & x < 1 \\ [x] \quad , & x \geq 1 \end{cases}$

For continuity at $x = 1$, we determine, $f(1)$, $\lim\limits_{x \to 1^-} f(x)$ and $\lim\limits_{x \to 1^+} f(x)$.

Now, $f(1) = [1] = 1$

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} \sin\frac{\pi x}{2} = \sin\frac{\pi}{2} = 1 \text{ and } \lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} [x] = 1$$

so $f(1) = \lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^+} f(x)$

$\therefore$ $f(x)$ is continuous at $x = 1$

2. $f(x) = \begin{cases} \frac{1}{2} - x & , \quad 0 \leq x < \frac{1}{2} \\ 1 & , \quad x = \frac{1}{2} \\ \frac{3}{2} - x & , \quad \frac{1}{2} < x \leq 1 \end{cases}$ at $x = \frac{1}{2}$

Ans discontinuous

Sol. : We have : $\left(\text{LHL at } x = \frac{1}{2}\right) = \lim\limits_{x \to \frac{1}{2}^-} f(x) = \lim\limits_{x \to \frac{1}{2}} \left(\frac{1}{2} - x\right) \left[\because f(x) = \frac{1}{2} - x \text{ for } 0 \leq x < \frac{1}{2}\right]$

$= \frac{1}{2} - \frac{1}{2} = 0$ [Using direct substitution method]

and $\left(\text{RHL at } x = \frac{1}{2}\right) = \lim\limits_{x \to \frac{1}{2}^+} f(x) = \lim\limits_{x \to \frac{1}{2}} \left(\frac{3}{2} - x\right) \left[\because f(x) = \frac{3}{2} - x \text{ for } \frac{1}{2} < x \leq 1\right]$

$= \frac{3}{2} - \frac{1}{2} = 1$ [Using direct substitution method]

$\Rightarrow \underset{x \to \frac{1}{2}^-}{\text{Lim}} \neq \underset{x \to \frac{1}{2}^+}{\text{Lim}}$

Hence $f(x)$ is not continuous at $x = \frac{1}{2}$. Clearly $f(x)$ is discontinuious at $x = \frac{1}{2}$

3. $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & \text{when } x \neq 3 \\ 6, & \text{when } x = 3 \end{cases}$ at $x = 3$.

Ans continuous

Sol. $f(3) = 6$ (given)

$\underset{x \to 3}{\text{Lim}} f(x) = \underset{x \to 3}{\text{Lim}} \frac{(x-3)(x+3)}{(x-3)} = 6$

$\because \underset{x \to 3}{\text{Lim}} f(x) = f(3)$

$\therefore$ $f(x)$ is continuous at $x = 3$.

Find k if f(x) is continuous (Q4 to Q7)

4. $f(x) \begin{cases} x^2 & x \leq 2 \\ x + k & x > 2 \end{cases}$ at $x = 2$

Ans. $k = 2$

Sol. $f(2^-) = f(2^+) = f(2)$

$4 = 2 + k = 4 \Rightarrow k = 2$

5. $f(x) \begin{cases} x + 1 & x \leq 1 \\ 6 - kx^2 & x > 1 \end{cases}$ at $x = 1$

Ans. $k = 4$

Sol. $f(1^-) = f(1^+) = f(1)$

$2 = 6 - k = 2 \Rightarrow k = 4$

6. $f(x) \begin{cases} \frac{\sin 5x - \sin x}{\sin 5x + \sin x} & x \neq 0 \\ k & x = 0 \end{cases}$ at $x = 0$

Ans. $k = \frac{2}{3}$

Sol. $f(0) = \lim_{x \to 0} f(x)$

$$k=\lim_{n\to 0}\frac{\sin 5x-\sin x}{\sin 5x+\sin x}\left(\frac{0}{0}\right)$$

$$k=\lim_{n\to 0}\frac{2\cos 3x\sin 2x}{2\sin 3x\cos 2x}=\frac{2}{3}$$

7. $f(x)\begin{cases}\dfrac{(e^x-1)^3\cos ec(3x)}{\ln(1+x^2)} & x\neq 0\\ k & x=0\end{cases}$ at $x=0$

Ans. $k=\dfrac{1}{3}$

Sol. $f(0)=\lim_{x\to 0}f(x)$

$$k=\lim_{n\to 0}\frac{\left(e^x-1\right)^3\cos ec^3x}{\ell n\left(1+x^2\right)}$$

$$k=\lim_{x\to 0}\left(\frac{\left(e^x-1\right)^3}{x^3}\right)\cdot\frac{x^3}{\left(\frac{\sin 3x}{3x}\right)\cdot 3x^3\left(\frac{\ell n\left(1+x^2\right)}{x^2}\right)}$$

$$k=\frac{1}{3}$$

8. Let $f(x)=\begin{cases}\dfrac{(e^{2x}+1)-(x+1)(e^x+e^{-x})}{x(e^x-1)} & \text{if } x\neq 0\\ k & \text{if } x=0\end{cases}$

if $f(x)$ is continuous at $x=0$ then k is equal to

(A) 1/2 (B) 1 (C) 3/2 (D) 2

Ans. B

Sol. $l=\lim_{x\to 0}\dfrac{(e^{2x}+1)-(e^x+e^{-x})-x(e^x+e^{-x})}{x^2\left(\dfrac{e^x-1}{x}\right)}$

multiply N^r and D^r by e^x

$$\lim_{x\to 0}\frac{e^x(e^{2x}+1)-(e^{2x}+1)-x(e^{2x}+1)}{x^2\cdot e^x}=\lim_{x\to 0}(e^{2x}+1)\left[\frac{(e^x-1)-x}{x^2}\right]=2\cdot\frac{1}{2}=1$$

9. Let $\alpha \in R$ be such that the function $f(x)=\begin{cases}\dfrac{\cos^{-1}(1-\{x\}^2)\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3} & ,x\neq 0\\ \alpha & ,x=0\end{cases}$ is continuous at $x=0$, where

$\{x\}=x-[x]$,$[x]$ is the greatest integer less than or equal to x. then find α.

Ans. No such α exists

Sol. $$f\left(0^-\right)=\lim_{x\to 0^-}\frac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\{x\}\left(1+\{x\}\right)\left(1-\{x\}\right)}$$

$$f\left(0^-\right)=\lim_{x\to 0^-}\frac{\cos^{-1}(0)}{1\cdot(1+1)}\cdot\left(\frac{\sin^{-1}\left(1-\{x\}\right)}{\left(1-\{x\}\right)}\right)$$

$$f\left(0^-\right)=\frac{\pi}{4}$$

$$f\left(0^+\right)=\lim_{x\to 0^+}\frac{\cos^{-1}\left(1-x^2\right)\sin^{-1}\left(1-x\right)}{x\left(1-x^2\right)}$$

$$=\lim_{x\to 0^+}\frac{\cos^{-1}\left(1-x^2\right)}{x}\cdot\frac{\pi}{2}$$

Let $\cos^{-1}(1-x^2)=\theta$

$1-x^2=\cos\theta$

$$x=\sqrt{1-\cos\theta}=\sqrt{2}\sin\frac{\theta}{2}$$

$$f\left(0^+\right)=\lim_{\theta\to 0^+}\frac{\theta}{\sqrt{2}\sin\frac{\theta}{2}}\cdot\frac{\pi}{2}=\frac{\pi}{\sqrt{2}}$$

$f(0^-)\neq f(0^+)$

No such α exists

1.3 CONTINUITY IN AN INTERVAL

1. In (a, b)

A function f is said to be continuous in (a, b) if f is continuous at each & every point $\in$(a, b)

2. In [a, b]

A function f is said to be continuous in a closed interval [a, b] if:

(i) f is continuous in the open interval (a, b),

(ii) f is right continuous at 'a'

(iii) f is left continuous at 'b'

Checkpoint for Continuity in an Interval

(i) Continuity of a function should be checked at the points where definition of a function changes.

(ii) Continuity of {f(x)} and [f(x)] should be checked at all points where f(x) becomes integer.

(iii) Continuity of sgn (f(x)) should be checked at the points where f(x) = 0

Q.1 For what value of a and b the function

$$f(x)=\begin{cases} x+a\sqrt{2}\sin x, & 0\le x<\pi/4 \\ 2x\cot x+b, & \pi/4\le x\le\pi/2 \\ a\cos 2x-b\sin x, & \pi/2<x\le\pi \end{cases}$$ is continuous in an interval $[0,\pi]$.

Ans $a=\pi/6$, $b=-\pi/12$

Sol. : $\because$ f(x) is continuous in an interval $[0,\pi]$

So it is also continuous at $x=\pi/4$, $x=\pi/2$.

$\therefore \quad \lim\limits_{x\to\pi/4^-} f(x) = \lim\limits_{x\to\pi/4^+} f(x)$

$\Rightarrow \quad \pi/4 + a = \pi/2 + b \quad ...(1)$

and $\lim\limits_{x\to\pi/2^-} f(x) = \lim\limits_{x\to\pi/2^+} f(x)$

$\Rightarrow \quad 0+b = -a- b \quad ...(2)$

Solving (1) and (2) $\Rightarrow a=\pi/6$, $b=-\pi/12$.

Q.2 $f(x)=\operatorname{sgn}(x-2)+\left\{x-\frac{1}{2}\right\}+[\ln x]$, check continuity in [1, 3].

Ans. Discontinuous at $x=\frac{3}{2},2,\frac{5}{2},e$

Sol. Let $g(x)=\operatorname{sgn}(x-2)$

$h(x)=\left\{x-\frac{1}{2}\right\}$

$P(x)=[\ell nx]$

x	g(x)	h(x)	P(x)	f(x)
2	DC	C	C	DC
3/2	DC	C	C	DC
5/2	DC	C	C	DC
1	DC	C	C	DC
e	DC	C	C	DC

Q.3 If $f(x)=\begin{cases}[\sin \pi x] & ; \quad 0 \le x < 1 \\ \left\{x-\frac{2}{3}\right\}\cdot \operatorname{sgn}\left(x-\frac{5}{4}\right) & ; \quad 1 \le x \le 2\end{cases}$, where $\{\,.\,\}$ represents fractional part function and $[\,.\,]$ is greatest integer function, then comment on the continuity of function in the interval $[0, 2]$.

Ans discontinuous at $x = \frac{1}{2}, 1\ \frac{5}{4}, \frac{5}{3}$

Sol. : (i) Continuity should be checked at the end-points of intervals of each definition i.e. $x = 0, 1, 2$

(ii) For $[\sin \pi x]$, continuity should be checked at all values of x at which $\sin \pi x \in I$

i.e. $x = 0, \frac{1}{2}$

(iii) For $\left\{x-\frac{2}{3}\right\}\cdot \operatorname{sgn}\left(x-\frac{5}{4}\right)$, continuity should be checked when $x - \frac{5}{4} = 0$

(as sgn (x) is discontinuous at $x = 0$) i.e. $x = \frac{5}{4}$ and when $x - \frac{2}{3} \in I$

i.e. $x = \frac{5}{3}$ (as $\{x\}$ is discontinuous when $x \in I$)

$\therefore$ overall discontinuity should be checked at $x = 0, \frac{1}{2}, 1, \frac{5}{4}, \frac{5}{3}$ and 2 check the discontinuity your self.

discontinuous at $x = \frac{1}{2}, 1\ \frac{5}{4}, \frac{5}{3}$

Q.4 $f(x)=\begin{cases}x^2+6 & x \in Q \\ 5x & x \notin Q\end{cases}$ find points where f(x) is continuous.

Ans. $x = 2, 3$

Sol. For continuous

$x^2 + 6 = 5x$

$x^2 - 5x + 6 = 0 \Rightarrow x = 2, 3$

Q.5 Let $f(x) = \lim\limits_{n\to\infty} \frac{x^{2n-1}+ax^2+bx}{x^{2n}+1}$. If $f(x)$ is continuous for all $x \in R$ then find the values of a and b.$(n \in N)$

Ans $a = 0$ and $b = 1$

Sol. $f(x)=\begin{cases}ax^2+bx & \text{for } -1 < x < 1 \\ \frac{a-b-1}{2} & x = -1 \\ \frac{a+b+1}{2} & x = 1 \\ \frac{1}{x} & \text{for } x > 1 \text{ or } x < -1\end{cases}$

for continuity at $x = 1$ we have $a + b = \frac{a+b+1}{2}$

hence, $a + b = 1$ ....(1)

for continuity at $x = -1$

$a - b = -1 \Rightarrow a - b = -1$ ....(2)

hence $a = 0$ and $b = 1$

Q.6 $f(x) \begin{cases} \lim\limits_{n\to\infty} \dfrac{x^n - \sin(x^n)}{x^n + \sin(x^n)} & , x > 0, x \neq 1 \\ 0 & , x = 1 \end{cases}$ check continuity.

Ans. Discontinuous at $x = 1$

Sol.
$$f(x) = \begin{cases} \lim\limits_{n\to\infty} \dfrac{1 - \dfrac{\sin x^n}{x^n}}{1 + \dfrac{\sin x^n}{x^n}} & x > 0, x \neq 1 \\ 0 & x = 1 \end{cases}$$

$$f(x) = \begin{cases} 0 & x \in (0,1) \\ 0 & x = 1 \\ 1 & x > 1 \end{cases}$$

f(x) is discontinuous at $x = 1$

Q7 Let $f(x) = \dfrac{\sqrt{x^2 + kx + 1}}{x^2 - k}$. The interval(s) of all possible values of k for which f is continuous for every $x \in R$, is

(A) $(-\infty, -2]$ (B) $[-2, 0)$ (C) $R - (-2, 2)$ (D) $(-2, 2)$

Ans B

Sol. $f(x) = \dfrac{\sqrt{x^2 + kx + 1}}{x^2 - k}$

for f to be continuous $\forall\ x \in R$

$x^2 + kx + 1 \geq 0$ and $x^2 - k$ must not have any root i.e. $k < 0$

$\therefore$ $k^2 - 4 \leq 0$ and $k < 0$ ....(1)

$\Rightarrow$ $k \in [-2, 2]$ ....(2)

from (1) and (2)

$k \in [-2, 0)$

2. TYPES OF DISCONTINUITY

Removable Type

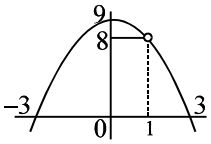
Here $\lim\limits_{x\to l} f(x)$ necessarily exists, but is either not equal to $f(a)$ or $f(a)$ is not defined.

In this case, therefore it is possible to redefine the function in such a manner that $\lim\limits_{x\to a} f(x) = f(a)$ and thus making the function continuous. Thesediscontinuities can be further classified as

(A) Missing point discontinuity

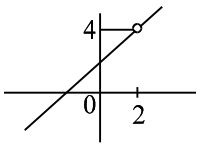
(a) $f(x)=\dfrac{(x-1)(9-x^2)}{x-1}$

at $x = 1$

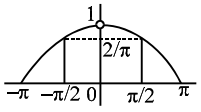


(b) $f(x)=\dfrac{x^2-4}{x-2}$

at $x = 2$



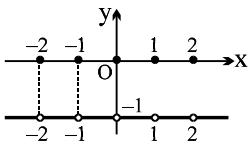
(c) $f(x)=\dfrac{\sin x}{x}$



(B) Isolated point discontinuity

(a) $f(x) = [x]+[-x]$

$$= \begin{cases} 0 & \text{if } x \in I \\ -1 & \text{if } x \notin I \end{cases}$$



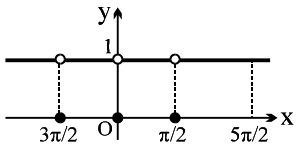
(b) $f(x)=\text{sgn}(\cos 2x-2\sin x+3)$

$= 2(2 + \sin x)(1 - \sin x)$

$$= \begin{cases} 0 & \text{if } x = 2n\pi+\dfrac{\pi}{2} \\ +1 & \text{if } x \neq 2n\pi+\dfrac{\pi}{2} \end{cases}$$

has an isolated point at $x = 0$

discontinuity as $x=2n\pi+\dfrac{\pi}{2}$



Non-Removable Type

Here $\lim\limits_{x\to a} f(x)$ does not exists and therefore it is not possible to redefine the function in any manner to make it continuous. Such discontinuities can be further classified into 3 fold.

(a) **Finite type** (both limits finite and unequal)

Examples of Finite type

(1) $\lim\limits_{x\to 0}\tan^{-1}\left(\dfrac{1}{x}\right)$ $\begin{cases} f(0^+)=\dfrac{\pi}{2} \\ f(0^-)=-\dfrac{\pi}{2} \end{cases}$; jump $= \pi$

(2) $\lim\limits_{x\to 0}\dfrac{|\sin x|}{x}$ $\begin{cases} f(0^+)=1 \\ f(0^-)=-1 \end{cases}$; jump $= 2$

(3) $\lim_{x\to 2}\frac{[x]}{x}$ $\begin{cases} f(2^+)=1 \\ f(2^-)=\frac{1}{2} \end{cases}$; jump $=\frac{1}{2}$

In this case non negative difference between the two limits is called the Jump of discontinuity.

(b) **Infinite type** (at least one of the two limit are infinity)

Examples of Infinite type

(1) $f(x)=\frac{x}{1-x}$ at $x=1$ $\begin{cases} f(1^+)=-\infty \\ f(1^-)=+\infty \end{cases}$

(2) $f(x)=2^{\tan x}$ at $x=\frac{\pi}{2}$ $\begin{cases} f\left(\frac{\pi^+}{2}\right)=0 \\ f\left(\frac{\pi^-}{2}\right)=\infty \end{cases}$

(3) $f(x)=\frac{1}{x^2}$ at $x=0$ $\begin{cases} f(0^+)=\infty \\ f(0^-)=\infty \end{cases}$

(c) **Oscillatory** (limits oscillate between two finite quantities)

Examples of Oscillatory

(1) $f(x)=\sin\frac{1}{x}$ or $f(x)=\cos\frac{1}{x}$ at $x=0$

Determine type of discontinuity for following functions

1. $f(x)=\frac{1}{\operatorname{cosec} x}$ at $x=0$

Ans. Missing point

Sol. $f(0^+)=\frac{1}{\infty}=0$

$f(0^-)=\frac{1}{-\infty}=0$

f(0) = Not differentiable

2. $f(x)=[x]+[-x]$ at $x=0$

Ans. Isolated point

Sol. $f(x)=\begin{cases} 0 & x\in I \\ -1 & x\notin I \end{cases}$

$f(0^+) = f(0^-) = -1$

But $f(0) = 0$

3. $f(x) = \text{sgn}(\sin x)$ at $x = 0$

Ans. Finite type

Sol. $f(0^+) = \text{sgn}(0^+) = 1$

$f(0^-) = \text{sgn}(0^-) = -1$

4. $f(x) = \text{sgn}(\sin^2 x)$ at $x = 0$

Ans. Isolated point

Sol. $f(0^+) = \text{sgn}(0^+) = 1$

$f(0^-) = \text{sgn}(0^-) = 1$

$f(0) = \text{sgn}(0) = 0$

5. $f(x) = 2^{\tan x}$ at $x = \dfrac{\pi}{2}$

Ans. Infinite type

Sol. $f\left(\dfrac{\pi^+}{2}\right) = 2^{-\infty} = 0$

$f\left(\dfrac{\pi^-}{2}\right) = 2^{\infty} = \infty$

6. $f(x) = e^{-\tan^2 x}$ at $x = \dfrac{\pi}{2}$

Ans. Missing point

Sol. $f\left(\dfrac{\pi^+}{2}\right) = e^{-\infty} = 0$

$f\left(\dfrac{\pi^-}{2}\right) = e^{-\infty} = 0$

$f\left(\dfrac{\pi}{2}\right) =$ Not differentiable

7. $f(x) = x \sin\left(\dfrac{1}{x}\right)$ at $x = 0$

Ans. Missing point

Sol. $f(0^+) = 0 \cdot \sin \infty = 0$

$f(0^-) = 0 \cdot \sin (1 - \infty) = 0$

$f(0) =$ Not differentiable

8. $f(x) = \dfrac{1}{|x|}$ at $x = 0$

Ans. Infinite type

Sol. $f\left(0^+\right) = \dfrac{1}{0^+} \to \infty$

$f\left(0^-\right) = \dfrac{1}{0^+} \to \infty$

Note :

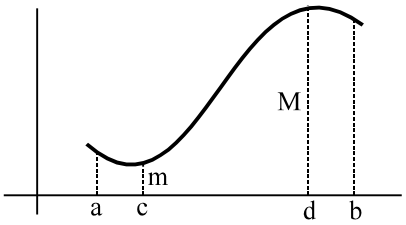
1. Function f(x) is discontinuous at x = a, b and g(x) is discontinuous at x = c, d then,

(i) Function (f + g) and (f – g) will be discontinuous at x = a, b, c and d.

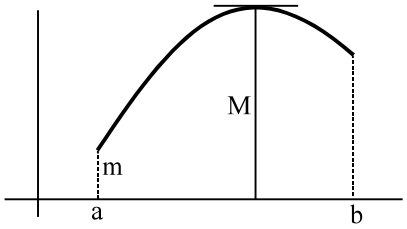
(ii) Check points for function (f × g) and $\left(\dfrac{f}{g}\right)$ will be x = a, b, c, d.

(iii) Check points for f(g(x)) will be : x = c, d , x when g(x) = a, x when g(x) = b

2. **Extreme Value Theorem :** If f is continuous on [a, b] then f takes on, a least value of m and a greatest value M on this interval.



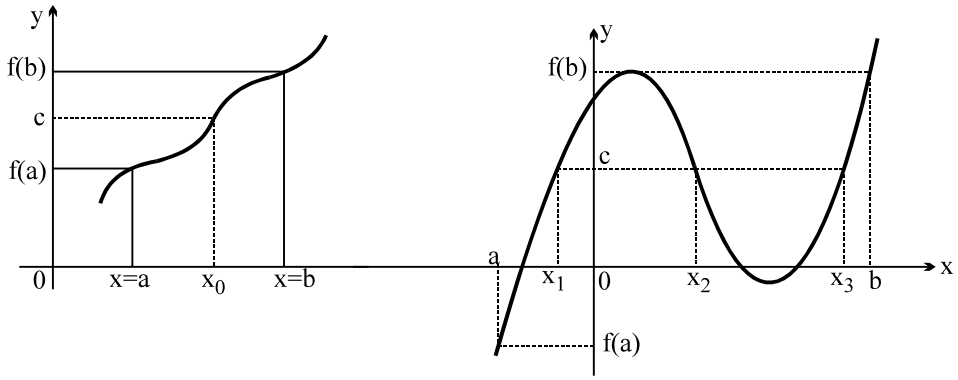
Minimum value 'm' occurs at x = c and maximum value M occurs at x = d. c, d ∈ (a, b)



Minimum value 'm' occurs at the end point x = a and the maximum value M occurs inside the interval

3. INTERMEDIATE VALUE THEOREM (IMVT)

If f is continuous on [a, b] and $f(a) \neq f(b)$ then for any value $c \in \left(f(a), f(b)\right)$, there is at least one number x_0 in (a, b) for which $f(x_0) = c$

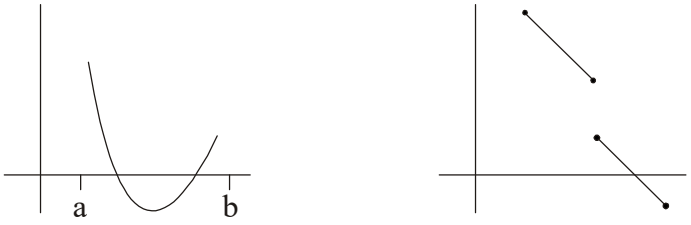


Alternate form of IMVT

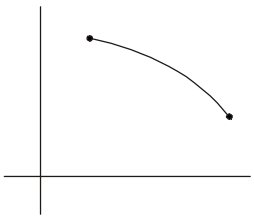
If f(x) is continuous in [a, b] and f(a) and f(b) are of opposite sign then there exists atleast one solution of equation f(x) = 0 in (a, b).

There can be more than one solution.

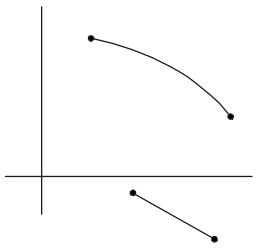
For equation f(x) = 0 to have a solution in (a, b) conditions of IMVT are sufficient but not necessary.



IMVT not applicable



IMVT ×



IMVT ×

Q.1 Prove that equation (x –1) (x –3) = λ (x –2) (x –4) will have atleast one solution in (2, 4)

Sol. Let f(x) = (x – 1) (x – 3) – λ(x – 2) (x – 4)

f(x) is continuous in (2, 4)

f(2) = (1) (–1) – 0 = –1 < 0

f(4) = 3·1 – 0 = 3 > 0

f(2)·f(4) < 0 ⇒ f(x) = 0 will have atleast one solution in (2, 4).

Q.2 Prove that equation $x^{2018}+\frac{1}{1+\sin^2 x}=2018$ will atleast one solution in (0, 2)

Sol. Let $f(x)=x^{2018}+\frac{1}{1+\sin^2 x}-2018$

f(x) is continuous in (0, 2)

f(0) = 0 + 1 – 2018 < 0

$f(2)=2^{2018}+\frac{1}{1+\sin^2 2}-2018>0$

$f(0)\cdot f(2)<0 \Rightarrow f(x)=0$ will have have atleast one solution in (0, 2)

Q.3 Prove that equation $(x-a)^4(x-b)^4+2x=a+b$ will have atleast one solution in (a, b)

Sol. Let $f(x)=(x-a)^4(x-b)^4+2x-(a+b)$

f(x) is continuous in (a, b)

f(a) = 0 + 2a – (a + b) = a – b

f(b) = 0 + 2b – (a + b) = b – a = – (a – b)

$f(a)\cdot f(b)<0 \Rightarrow f(x)=0$ will have atleast one solution in (a, b).

Q.4 f: [0, 1] onto [0, 1] is a continuous function then prove that there exists at least one c in [0, 1] such that $f(c)=1-c$

Sol. Consider $g(x)=f(x)-1+x$

$g(0)=f(0)-1 \le 0$ (as $f(0)\le 1$)

$g(1)=f(1)\ge 0$ (as $f(1)\ge 0$)

Hence g (0) and g (1) and of opposite signs

hence there exist at least one $c \in [0, 1]$ such that $g(c)=0$

$\therefore\ g(c)=f(c)-1+c=0$; $f(c)=1-c$

Q.5 Let $f:[0,1]\to R$ is a continuous function such that f(0) = f(1). Prove that there exists atleast one c in $\left[0,\frac{1}{2}\right]$ such that $f(c)=f\left(c+\frac{1}{2}\right)$

Sol. Let $g(x)=f(x)-f\left(x+\frac{1}{2}\right)$

g(x) is continuous in $\left[0,\frac{1}{2}\right]$

$g(0)=f(0)-f\left(\frac{1}{2}\right)$

$$g\left(\frac{1}{2}\right)=f\left(\frac{1}{2}\right)-f(1)=f\left(\frac{1}{2}\right)-f(0)\left(f(0)=f(1)\right)$$

$g(0)\cdot g\left(\frac{1}{2}\right)\le 0 \Rightarrow$ there exist $\exists c\in\left[0,\frac{1}{2}\right]$

such that $g(c)=0$

$$\Rightarrow f(c)-f\left(c+\frac{1}{2}\right)=0$$

$$\Rightarrow f(c)=f\left(c+\frac{1}{2}\right)$$

Q.6 Let $f:[0,1]\to R$ is a continuous function such that $f(0)=f\left(\frac{1}{2}\right)=f(1)$ then prove that equation $f(x)=f\left(x+\frac{1}{4}\right)$ have atleast 2 solutions in $\left[0,\frac{3}{4}\right]$.

Sol. Let $g(x)=f(x)-f\left(x+\frac{1}{4}\right)$

$g(x)$ is continuous in $[0,1]$

$$g(0)=f(0)-f\left(\frac{1}{4}\right)$$

$$g\left(\frac{1}{4}\right)=f\left(\frac{1}{4}\right)-f\left(\frac{1}{2}\right)=f\left(\frac{1}{4}\right)-f(0) \qquad \left(f(0)=f\left(\frac{1}{2}\right)\right)$$

$g(0)\cdot g\left(\frac{1}{4}\right)\le 0 \Rightarrow$ there exists $\exists c\in\left[0,\frac{1}{4}\right]$

such that $g(c_1)=0$

again $g\left(\frac{1}{2}\right)=f\left(\frac{1}{2}\right)-f\left(\frac{3}{4}\right)$

$$g\left(\frac{3}{4}\right)=f\left(\frac{3}{4}\right)-f(1)=f\left(\frac{3}{4}\right)-f\left(\frac{1}{2}\right) \qquad \left(f\left(\frac{1}{2}\right)=f(1)\right)$$

$g\left(\frac{1}{2}\right)\cdot g\left(\frac{3}{4}\right) \le 0 \Rightarrow$ there exists $\exists c_2 \in \left[\frac{1}{2}, \frac{3}{4}\right]$

s.t. $g(c_2) = 0$

$\Rightarrow$ there exists two $c_1, c_2 \in \left[0, \frac{3}{4}\right]$ where $g(x) = 0$

Q.7 Let f : [0, 2] $\to$ R be continuous and f (0) = f (2). Prove that there exists x_1 and x_2 in (0, 2) such that $x_2 - x_1 = 1$ and $f(x_2) = f(x_1)$

Sol. Consider continuous function g as

$g(x) = f(x + 1) - f(x)$ $\quad (x_2 = x_1 + 1)$

now, $g(0) = f(1) - f(0) = f(1) - f(2)$ ....(1)

$g(1) = f(2) - f(1) = f(2) - f(1)$ ....(2)

hence g (0) and g (1) are of opposite signs, hence $\exists$ some $c \in (0, 1)$ where g (c) = 0

i.e. $f(c + 1) = f(c)$ $\quad [c + 1 \in (1, 2)$ as $c \in (0, 1)\ ]$

put $c = x_1$; $c + 1 = x_2$

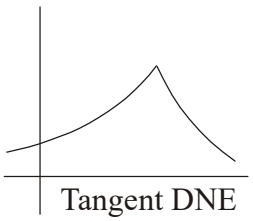
$\therefore$ $f(x_2) = f(x_1)$ where $x_2 - x_1 = 1$

obviously $x_1, x_2 \in (0, 2)$

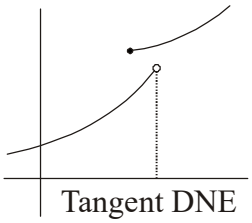
DIFFERENTIABILITY

1.1 DEFINITION OF DIFFERENTIABILITY

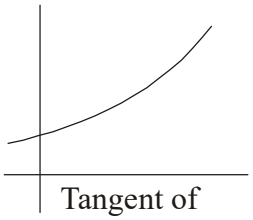
Geometrical Meaning of Derivative



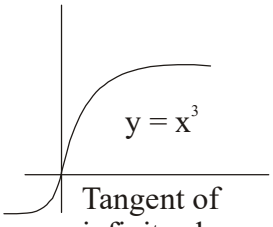
Tangent DNE



Tangent DNE



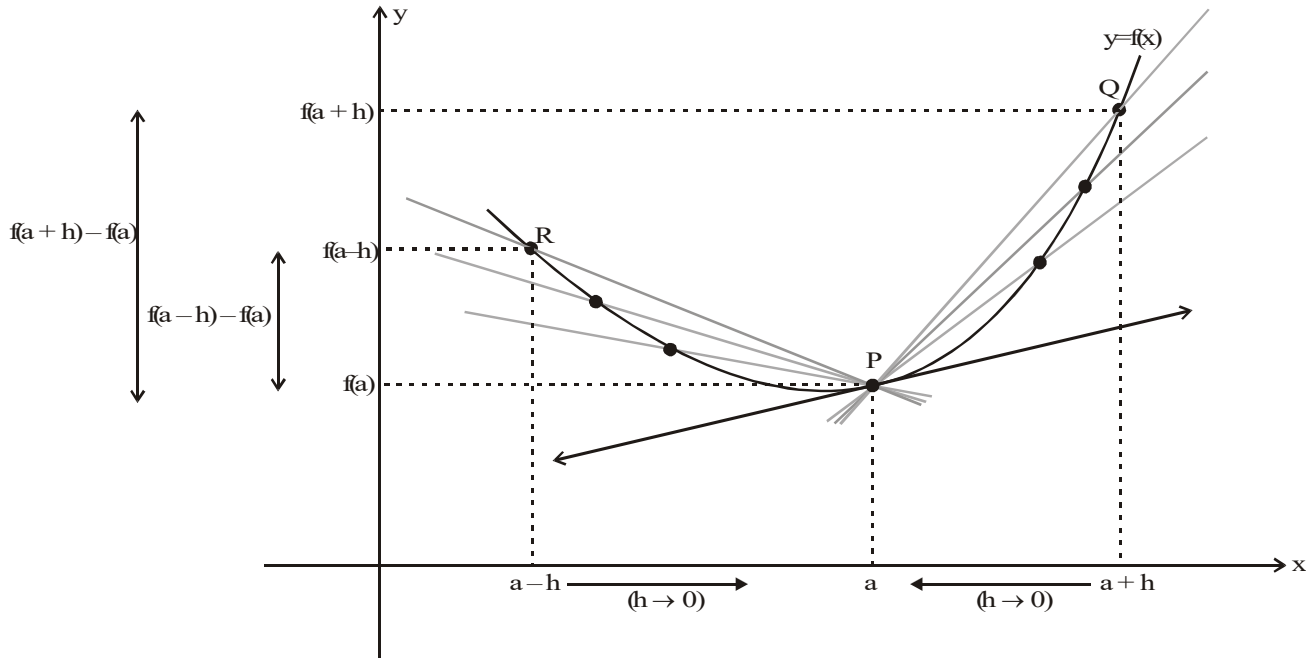
Tangent of finite slope exists



Tangent of infinite slope exists.

If at x = a tangent of finite slope exists for function f(x) then we say that function is derivable at x = a.

Existence of derivative at x = a :



(a) Right hand derivative :

The right hand derivative of f(x) at x = a denoted by $f'(a^+)$ is defined as :

$$f'(a^+) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$, provided the limit exists & is finite. (h > 0)

(b) Left hand derivative :

The left hand derivative of f(x) at x = a denoted by $f'(a^-)$ is defined as :

$$f'(a^-) = \lim_{h\to 0} \frac{f(a-h)-f(a)}{-h}$$, provided the limit exists & is finite. (h > 0)

Hence f(x) is said to be derivable or differentiable at x = a.

If $f'(a^+) = f'(a^-)$ = finite quantity

and it is denoted by f'(a); where $f'(a) = f'(a^-) = f'(a^+)$

Check differentiability of following functions

1. $f(x)$ $\begin{cases} 1+\sin x & x \geq 0 \\ \cos x & x < 0 \end{cases}$ at x = 0

Ans. LHD = 0, RHD = 1

Sol. $$\text{LHD} = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0} \frac{\cos(-h)-1}{(-h)} = 0$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{(1+\sinh)-1}{h} = 1$$

Not differentiable at $x = 0$

2. $f(x)$ $\begin{cases} e^x - 1 & x > 0 \\ \sin x & x \le 0 \end{cases}$ at $x = 0$

Ans. LHD = 1, RHD = 1

Sol. $$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}\frac{\sin(-h)}{(-h)} = 1$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{(e^h-1)-0}{h} = 1$$

Differentiable at $x = 0$

3. $f(x) = [x]$ at $x = 1$

Ans. LHD = ∞, RHD = 0

Sol. $$LHD = \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0}\frac{0-1}{-h} \to \infty$$

$$RHD = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{1-1}{-h} = 0$$

Not differentiable at $x = 1$

4. $f(x) = \text{sgn}(x)$ at $x = 0$

Ans. LHD = ∞, RHD = ∞

Sol. $$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}\frac{-1-0}{-h} \to \infty$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{1-0}{h} \to \infty$$

Not differentiable at $x = 0$

5. $f(x) = |x|$ at $x = 0$

Ans. LHD = –1, RHD = 1

Sol. $$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}\frac{|-h|-0}{-h} = \frac{h}{-h} = -1$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{|h|-0}{h} = \frac{h}{h} = 1$$

Not differentiable at x = 0

6. $f(x) = e^{|x|}$ at x = 0

Ans. LHD = – 1, RHD = 1

Sol. $$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}\frac{e^{|-h|}-1}{-h} = \lim_{h\to 0}\frac{e^{h}-1}{-h} = -1$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{e^{|h|}-1}{h} = \lim_{h\to 0}\frac{e^{h}-1}{h} = 1$$

Not differentiable at x = 0

7. $f(x) = |\ln x|$ at x = 1

Ans. LHD = –1, RHD = 1

Sol. $$LHD = \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0}\frac{|\ell n(1-h)|-0}{-h} = \lim_{h\to 0}\frac{-\ell n(1-h)}{-h} = -1$$

$$RHD = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{|\ell n(1+h)|-0}{h} = \lim_{h\to 0}\frac{\ell n(1+h)}{h} = 1$$

Not differentiable at x = 1

8. $f(x) = x|x|$ at x = 0

Ans. LHD = 0, RHD = 0

Sol. $$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}\frac{-h|-h|-0}{-h} = 0$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{h|h|-0}{h} = 0$$

Differentiable at x = 0

9. $f(x) = [x^2]$ at x = 0

Ans. LHD = 0, RHD = 0

Sol. $$LHD = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}\frac{\left[(-h)^2\right]-0}{-h} = \frac{0-0}{-h} = 0$$

$$RHD = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{\left[h^2\right]-0}{h} = \frac{0-0}{h} = 0$$

Differentiable at x = 0

10. $f(x) = \ln^2 x$ at x = 1

Ans. LHD = 0, RHD = 0

Sol. $$LHD = \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0}\frac{\ell n^2(1-h)-0}{(-h)} = 0$$

$$RHD = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{\ell n^2(1+h)-0}{h} = 0$$

Differentiable at x = 0

11. $f(x) = \cos x + |\cos x|$ at $x = \frac{\pi}{2}$

Ans. LHD = –2, RHD = 0

Sol. $$LHD = \lim_{h\to 0}\frac{f\left(\frac{\pi}{2}-h\right)-f\left(\frac{\pi}{2}\right)}{-h} = \lim_{h\to 0}\frac{\sinh+|\sinh|-0}{-h} = -2$$

$$RHD = \lim_{h\to 0}\frac{f\left(\frac{\pi}{2}+h\right)-f\left(\frac{\pi}{2}\right)}{h} = \lim_{h\to 0}\frac{-\sinh+|-\sinh|-0}{h} = 0$$

Not differentiable at $x = \frac{\pi}{2}$

1.2 IMPORTANT POINTS

1. If a function is derivable at x = a, then x = a must be in domain of function.
2. Differentiability $\Rightarrow$ Continuity ; Continuity $\not\Rightarrow$ derivability ;
3. Non derivibality $\not\Rightarrow$ discontinuous But discontinuity $\Rightarrow$ Non derivability
4. If RHD = a finite no then function is said to be right derivable at x = a.
5. If LHD = a finite no then function is said to be left derivable at x = a.
6. All Polynomials, Trigonometrical functions, exponential and Logarithmic functions are derivable in their domain .

7. If f(x) and g(x) both are derivable at x = a, then function (f + g), (f – g), (f × g) and $\left(\frac{f}{g}\right)$ will be derivable at x = a.

8. If f(x) is derivable at x = a and g(x) is non-derivable at x = a then,

   (i) function (f + g) and (f – g) will be non-derivable at x = a.

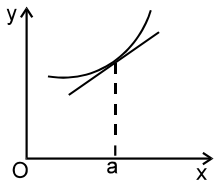
   (ii) Nothing definite can be said about derivability of (f × g) and $\left(\frac{f}{g}\right)$ at x = a.

9. If both functions f(x) and g(x) are non-derivable at x = a, then nothing definite can be said about derivability of (f + g), (f – g), (f × g) and $\left(\frac{f}{g}\right)$ at x = a.

1.3 CONCEPT OF LHD AND RHD

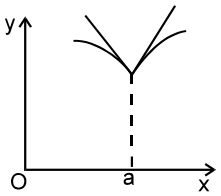
If f(x) is a function such that R.H.D = $f'(a^+) = \ell$ and L.H.D. = $f'(a^-) = m$.

1. If $\ell = m$ = some finite value, then the function f(x) is differentiable as well as continuous.
2. if $\ell \neq m$ = but both have some finite value, then the function f(x) is non differentiable but it is continuous.
3. If at least one of the ℓ or m is infinite, then the function is non differentiable but we can not say about continuity of f(x).



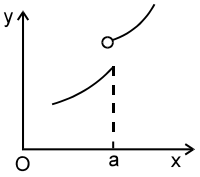
(i)

continuous and differentiable



(ii)

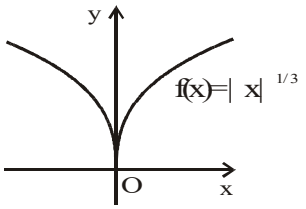
continuous but not differentiable



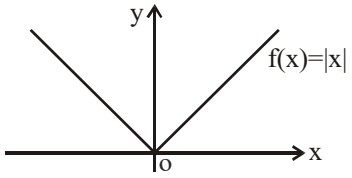
(iii)

neither continuous nor differentiable

f(x)= |x| and $|x|^{1/3}$ is continuous but not differentiable at x = 0 & there is sharp corner at x = 0.

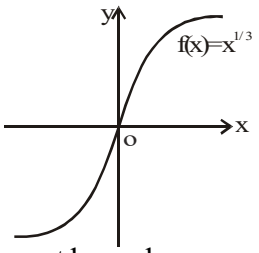


(does not have unique tangent) $\begin{cases} p \to +\infty \\ q \to -\infty \end{cases}$



(does not have unique tangent) $\begin{cases} p = 1 \\ q = -1 \end{cases}$

$f(x) = x^{1/3}$ is continuous but not differentiable at x = 0 because $f'(0^+) \to \infty$ and $f'(0^-) \to \infty$.



(unique tangent but does not have sharp corner) $p \to +\infty$, $q \to +\infty$

Note : sharp corner $\Rightarrow$ non differentiable | non differentiable $\not\Rightarrow$ sharp corner

2.1 DIFFERENTIABILITY IN AN INTERVAL

1. In (a, b)

f(x) is said to be differentiable over an open interval (a, b) if it is differentiable at each & every point of the open interval (a, b)

2. In [a, b]

f(x) is said to be differentiable over the closed interval [a, b] if :

(i) f(x) is differentiable in (a, b)

(ii) It is right derivable at x = a.

(iii) It is left derivable at x = b.

Checkpoint for Differentiability in an Interval

(i) At all points where continuity is required to be checked.

(ii) |f(x)| ; points where f(x) = 0.

Q.1 $$f(x)\begin{cases} |x-3| & x \geq 1 \\ \dfrac{x^2}{4}-\dfrac{3x}{2}+\dfrac{13}{4} & x < 1 \end{cases}$$

Ans. Non-derivable at x = 3

Sol. $$f(x)=\begin{cases} x-3 & x \geq 3 \\ 3-x & x \in [1,3) \\ \dfrac{x^2}{4}-\dfrac{3x}{2}+\dfrac{13}{4} & x < 1 \end{cases}$$

Check points $\Rightarrow$ x = 1, 3

at x = 1

$$\text{LHD} = \lim_{h\to 0}\frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0}\frac{\left(\frac{(1-h)^2}{4}-\frac{3(1-h)}{2}+\frac{13}{4}\right)-2}{-h} = \lim_{h\to 0}\frac{h^2+4h}{-4h} = -1$$

$$\text{RHD} = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{h\to 0}\frac{(3-(1+h))-2}{h} = -1$$

Differentiable at x = 1

at x = 3

$$\text{LHD} = \lim_{h\to 0}\frac{f(3-h)-f(3)}{-h} = \lim_{h\to 0}\frac{(3-(3-h))-0}{-h} = -1$$

$$\text{RHD} = \lim_{h\to 0}\frac{f(3+h)-f(3)}{h} = \lim_{h\to 0}\frac{(3+h-3)-0}{h} = 1$$

Not differentiable at x = 3

Q.2 If $f(x) = \begin{cases} \left\{x+\frac{1}{3}\right\}[\sin \pi x] & , \ 0 \le x < 1 \\ [2x]\,\text{sgn}\left(x-\frac{4}{3}\right) & , \ 1 \le x \le 2 \end{cases}$, find those points at which continuity and differentiability should be checked, where [.] is greatest integer function and {.} is fractional part function.

Also check the continuity and differentaibility of f(x) at x = 1.

Ans $x = 0, \frac{1}{2}, \frac{2}{3}, 1, \frac{4}{3}, \frac{3}{2}, 2$; discontinuous and non diffferentiable at x = 1.

Sol. $f(x) = \begin{cases} \left\{x+\frac{1}{3}\right\}[\sin \pi x] & , \ 0 \le x < 1 \\ [2x]\,\text{sgn}\left(x-\frac{4}{3}\right) & , \ 1 \le x \le 2 \end{cases}$ The points, where we should check the continuity and differentiability are $x = 0, \frac{1}{2}, \frac{2}{3}, 1, \frac{4}{3}, \frac{3}{2}, 2$

At x = 1

$$\text{L.H.L.} = \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \left\{x+\frac{1}{3}\right\}[\sin \pi x] = 0$$

R.H.L. $= \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} [2x] \operatorname{sgn}\left\{x-\frac{4}{3}\right\} = 2(-1) = -2$

$\therefore$ L.H.L $\neq$ R.H.L. hence f(x) is discontinuous at x = 1 and hence it is non diffferentiable at x = 1.

Q.3 If $f(x)=\begin{cases} e^{-|x|}, & -5<x<0 \\ -e^{-|x-1|}+e^{-1}+1, & 0\le x<2 \\ e^{-|x-2|}, & 2\le x<4 \end{cases}$

Discuss the continuity and differentiability of f(x) in the interval (–5, 4).

Ans Always Continuous but not differentiable at x = 0,1,2

Sol. : Check the differentiability at x = 0

$$LHD=\lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}=\lim_{h\to 0}\frac{e^{-|-h|}-1}{-h}=\lim_{h\to 0}\frac{e^{-h}-1}{-h}=1$$

$$RHD=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}=\lim_{h\to 0}\frac{-e^{-|h-1|}+e^{-1}+1-1}{h}=\lim_{h\to 0}\frac{e^{-1}(1-e^{h})}{h}=-e^{-1}$$

$\therefore$ LHD $\neq$ RHD

$\Rightarrow$ not differentiable at x = 0.

but f(x) is continous at x = 0, because p $\neq$ q and both are finite.

check the differentiability at x = 2

$$LHD=\lim_{h\to 0}\frac{f(2-h)-f(2)}{-h}=\lim_{h\to 0}\frac{e^{-|1-h|}+e^{-1}+1-1}{-h}=\lim_{h\to 0}\frac{e^{-1}(1-e^{h})}{-h}=e^{-1}$$

$$RHD=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=\lim_{h\to 0}\frac{e^{-|h|}-1}{h}=\lim_{h\to 0}\frac{(e^{-h}-1)}{h}=-1$$

$\therefore$ LHD $\neq$ RHD

$\Rightarrow$ not differentiable at x = 2.

but f(x) is continous at x = 2, because p $\neq$ q and both are finite.

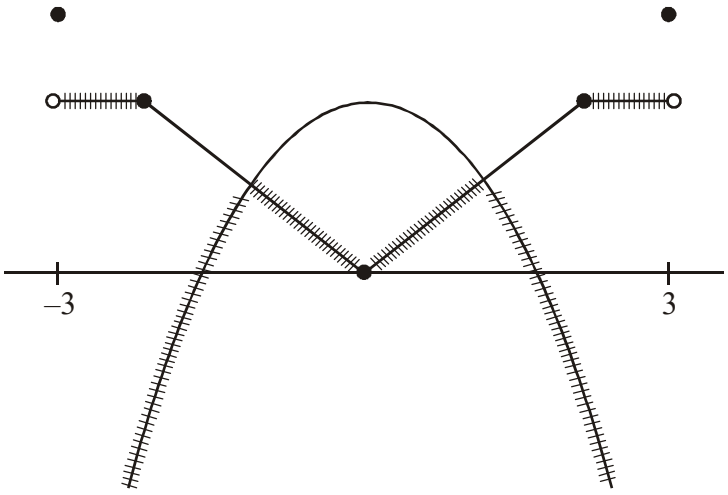
Q 4. A function f is defined on $[-3, 3]$ as

$$f(x)=\begin{cases}\min\{|x|, 2-x^2\}, & -2 \le x \le 2 \\ [|x|], & 2<|x| \le 3\end{cases}$$

where $[x]$ denotes the greatest $r \le x$. The number of points, where f is not differentiable in $(-3, 3)$ is :

Ans. 5

Sol.



not differentiable at 5 point

Q. 5. Let $[x]$ denote the greatest integer function and $f(x) = \max\{1+x+[x], 2+x, x+2[x]\}, 0 \le x \le 2$. Let m be the number of points in $[0,2]$, where f is not continuous and n be the number of points in $(0, 2)$, where f is not differentiable. Then $(m+n)^2+2$ is equal to :

यदि $[x]$ महत्तम पूर्णांक फलन है तथा $f(x) = \max\{1+x+[x], 2+x, x+2[x]\}, 0 \le x \le 2$ है। यदि $[0,2]$ में बिन्दुओं की संख्या m इस प्रकार है जहाँ f असतत् है तथा $(0, 2)$ में बिन्दुओं की संख्या इस प्रकार है जहाँ f अवकलनीय नहीं है, तो $(m+n)^2+2$ का मान बराबर होगा :

Ans. 3

Sol. $$f(x)=\begin{cases}\max\{x+1, x+2, x\} & 0 \le x < 1 \\ \max\{x+2, x+2, x+2\} & 1 \le x < 2 \\ \max\{5,4,6\} & x = 2\end{cases}$$

$$\Rightarrow f(x)=\begin{cases}x+2 & 0 \le x < 1 \\ x+2 & 1 \le x < 2 \\ 6 & x = 2\end{cases}$$

$$\Rightarrow f(x)=\begin{vmatrix}x+2 & 0 \le x < 2 \\ 6 & x = 2\end{vmatrix}$$

$$\Rightarrow f(x)=\begin{cases} x+2 & 0 \le x < 2 \\ 6 & x = 2 \end{cases}$$

f is not continuous at $x = 2$
f is differentiable in (0, 2)
$\therefore\ m = 1, n = 0$
$(m + n)^2 + 2 = 1 + 2 = 3$

2.2 QUICK CONCEPT

(i) Function should be continuous at that point.

(ii) Differentiation of function should be simple.

(iii) Not applicable in oscillatory limits.

$$f(x)=\begin{cases} |x-3| & x \ge 1 \\ \dfrac{x^2}{4}-\dfrac{3x}{2}+\dfrac{13}{4} & x < 1 \end{cases}$$

Sol.

$$f(x)=\begin{cases} x-3 & x \ge 3 \\ 3-x & 1 \le x < 3 \\ \dfrac{x^2}{4}-\dfrac{3x}{2}+\dfrac{13}{4} & x < 1 \end{cases}$$

Check continuity at $x = 1$ and 3.

at $x = 1$	at $x = 3$
LHL = 2	LHL = 0
RHL = 2	RHL = 0
cont.	cont.

$$f'(x)=\begin{cases} 1 & x > 3 \\ -1 & 1 < x < 3 \\ \dfrac{x}{2}-\dfrac{3}{2} & x \le 1 \end{cases}$$

at $x = 1$	at $x = 3$
$LHD = f'(1^-) = \dfrac{1}{2}-\dfrac{3}{2} = -1$	$LHD = -1$
$RHD = f'(1^+) = -1$	$RHD = 1$

Non differentiable

2. $f(x)\begin{cases} e^x & x \geq 0 \\ 1+x & x < 0 \end{cases}$

Ans. always derivable

Sol. at $x = 0$

$$\left.\begin{aligned} LHL &= 1+0=1 \\ RHL &= e^0 = 1 \\ f(0) &= e^0 = 1 \end{aligned}\right\} \Rightarrow (c) \text{ at } x = 0$$

$$f'(x) = \begin{cases} e^x & x \geq 0 \quad f'(0^-) = 1 \\ 1 & x < 0 \quad f'(0^+) = e^0 = 1 \end{cases}$$

$\Rightarrow$ (D)

at $x = 0$

3. $f(x)\begin{cases} e^x - e + 9 & x \geq 1 \\ x^{100} + 7x^{50} + 1 & x < 1 \end{cases}$

Ans. Non-derivable at $x = 1$

Sol. at $x = 1$

LHL = 9, RHL = 9, $f(1) = 9 \Rightarrow$ (c) at $x = 1$

$$f'(x) = \begin{cases} e^x & x \geq 1 \quad f'(1^-) = 450 \\ 100x^{99} + 350x^{49} & x < 1 \quad f'(1^+) = e \end{cases}$$

Not differentiable at $x = 1$

4. $f(x)\begin{cases} -\cos x & -\pi/2 < x \leq 0 \\ x - 1 & 0 < x \leq 1 \\ \ln x & x > 1 \end{cases}$

Ans. Non-derivable at $x = 0$

Sol. at $x = 0$, LHL = –1, RHL = –1, $f(0) = -1 \Rightarrow$ (c) at $x = 0$

at $x = 1$, LHL = 0, RHL = 0, $f(1) = 0 \Rightarrow$ (c) at $x = 1$

$$f'(x)=\begin{cases}\sin x & -\frac{\pi}{2}<x\le 0\\ 1 & 0<x\le 1\\ \frac{1}{x} & x>1\end{cases}$$

$$\left.\begin{matrix}f'(0^-)=0\\ f'(0^+)=1\end{matrix}\right\}\Rightarrow \text{Not differentiable at } x=0$$

$$\left.\begin{matrix}f'(1^-)=1\\ f'(1^+)=1\end{matrix}\right\}\Rightarrow \text{Differentiable at } x=1$$

5. $f(x)\begin{cases}\rightarrow ax+b & x\le -1\\ \rightarrow ax^5+x+2b & x>-1\end{cases}$ find a and b if f(x) is derivable.

Ans. $a=\frac{-1}{4}, b=1$

Sol. first we need to nake f(x)
continuous at $x=-1$
$f(-1^-)=f(-1^+)=f(-1)$
$-a+b=-a-1+2b=-a+b\Rightarrow b=1$
Now,

$$f'(x)=\begin{cases}a & x\le -1\\ 5ax^4+1 & x>-1\end{cases}$$

$$f'(-1^+)=f'(-1^-)$$

$$5a+1=a\Rightarrow a=-\frac{1}{4}$$

Note:
1. Derivative of a differentiable function need not be a continuous function

Eg. $f(x)=\begin{cases}x^2\sin\frac{1}{x} & \text{if } x\ne 0\\ 0 & \text{if } x=0\end{cases}$; $f'(0^+)=0$, $f'(0^-)=0$

$$f'(x) = \begin{cases} \sin\frac{1}{x}\cdot 2x - x^2\cos\frac{1}{x}\left(\frac{1}{x^2}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$f'(x)$ is not continuous at $x = 0$.

2. $h(x) = |f(x)|$; $f(x)$ is differentiable

Case I	Case II
$f(a) \neq 0$. $\Rightarrow h(x)$ is derivable at $x = a$.	$f(a) = 0$. $\Rightarrow$ we will check derivability at $x = a$.

3. $h(x) = g(x)\,|f(x)|$; g and f are differentiable, $f(a) = 0$.

$g(a) = 0$	$g(a) \neq 0$
$\Rightarrow h$ is derivable at $x = a$.	$\Rightarrow$ we will check derivability at $x = a$.

$$h(x) \begin{cases} \longrightarrow g(x)\cdot f(x) & x \geq a \\ \longrightarrow -g(x)\cdot f(x) & x < a \end{cases}$$

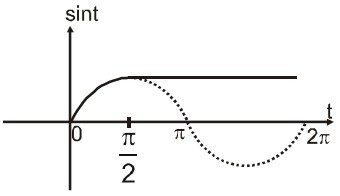
Proof:
$$h'(x) \begin{cases} \longrightarrow g'(x)\cdot f(x) + f'(x)\,g(x) & x > a \\ \longrightarrow -g'(x)\cdot f(x) - f'(x)\cdot g(x) & x < a \end{cases}$$

3. DEFINING SPECIAL TYPES OF FUNCTIONS

Q.1 $f(x) = \max(\sin t, 0 \leq t \leq x)$, $0 \leq x \leq 2\pi$ discuss the continuity of this function at $x = \frac{\pi}{2}$

Solution : $f(x) = \text{maximum}(\sin t, 0 \leq t \leq x)$, $0 \leq x \leq 2\pi$

if $x \in \left[0, \frac{\pi}{2}\right]$, $\sin t$ is increasing function



Hence if $t \in [0, x]$, $\sin t$ will attain its maximum value at $t = x$.

$\therefore f(x) = \sin x$ if $x \in \left[0, \frac{\pi}{2}\right]$

if $x \in \left(\frac{\pi}{2}, 2\pi\right]$ and $t \in [0, x]$

then sin t will attain its maximum value when $t = \frac{\pi}{2}$

$\therefore f(x) = \sin\frac{\pi}{2} = 1$ if $x \in \left(\frac{\pi}{2}, 2\pi\right]$ $\therefore f(x) = \begin{cases} \sin x & , \text{ if } x \in \left[0, \frac{\pi}{2}\right] \\ 1 & , \text{ if } x \in \left(\frac{\pi}{2}, 2\pi\right] \end{cases}$

Now $f\left(\frac{\pi}{2}\right) = 1$

$\lim\limits_{x \to \frac{\pi}{2}^-} f(x) = \lim\limits_{x \to \frac{\pi}{2}^-} \sin x = 1$

$\lim\limits_{x \to \frac{\pi}{2}^+} f(x) = \lim\limits_{x \to \frac{\pi}{2}^+} 1 = 1$

as $f(\pi/2)$ = L.H.S. = R.H.S. $\therefore$ f(x) is continuous at $x = \frac{\pi}{2}$

Q.2 $f(x) = \min(\sin t, 0 \le t \le x)$, $x > 0$.

Q.3 $f(x) = \min(\cos t, 0 \le t \le x)$, $x > 0$.

Q.4 $f(x) = \max(t^3 - 3t + 2, -2 \le t \le x)$; $x > -2$

4. DETERMINATION OF FUNCTION

1. Function f(x) is a differentiable function satisfying $f(x + y) = f(x) + f(y)\ \forall\ x, y \in R$, $f'(0) = 2$
find f(x)

BASIC STEPS

(1) Write down the expression for f' (x) as $f'(x) = \frac{f(x+h) - f(x)}{h}$

(2) Manipulate f(x + h) – f(x) in such a way that the given functional rule is applicable. Now apply the functional rule and simplify the RHS to get f' (x) as a function of x along with constants if any.

(3) Integrate f' (x) to get f(x) as a function of x and a constant of integration. In some cases a Differential Equation is formed which can be solved to get f(x).

(4) Apply the boundary value conditions to determine the value of this constant.

Sol $f(x + y) = f(x) + f(y)$

$f'(x) = \lim\limits_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim\limits_{h \to 0} \frac{f(x) + f(h) - f(x)}{h}$

$= \lim\limits_{h \to 0} \frac{f(0+h) - f(0)}{h}$ (as f (0) = 0)

$f'(x) = f'(0)$

$f'(x) = 2$

Integrate wrt x

$f(x) = 2x + c$

Another method :

$f(x + y) = f(x) + f(y)$

Partially diff w.r.t. x $\quad f'(x + y) = f'(x) + 0$

Partially diff w.r.t. y $\quad f'(x + y) = f'(y)$

$\Rightarrow \quad f'(x) = f'(y)$

put $y = 0$

$f'(x) = f'(0)$

$f'(x) = 2.$

Q.2 Function f(x) is a differentiable function satisfying $f(x + y) = f(x) + f(y) + 2xy - 1 \ \forall \ x, y \in R$, $f'(0) = 4$ find f(x).

Ans $f(x) = x^2 + 4x + 1$

Sol $$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{f(x) + f(h) + 2xh - 1 - f(x)}{h}$$

$$= 2x + \lim_{h\to 0} \frac{f(h) - 1}{h} \qquad [\text{Put } x = 0, y = 0 \text{ to get } f(0) = 1]$$

$$= 2x + \lim_{h\to 0} \frac{f(h) - f(0)}{h}$$

$f'(x) = 2x + f'(0)$

integrating $\quad f(x) = x^2 + f'(0)x + c$

if $\quad x = 0 \ ; f(0) = 1 \ \Rightarrow \ c = 1$

Q.3 Function f(x) is a differentiable function satisfying $f(x + y) = f(x).\ f(y) \ \forall \ x, y \in R$, $f'(0) = 2$, $f(0) \neq 0$ find f(x).

Ans $f(x) = e^{2x}$

Sol $$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{f(x)\cdot f(h) - f(x)}{h} = \lim_{h\to 0} f(x) \frac{[f(h) - 1]}{h}$$

$[\text{Put } x = 0, y = 0; f(0) = f^2(0) \Rightarrow f(0) = 0 \text{ or } f(0) = 1]$

$$= f(x) \frac{[f(h) - f(0)]}{h} = f(x) \cdot f'(0) \quad [\text{if } f(0) = 0 \text{ then } f(x) = 0]$$

$$\frac{f'(x)}{f(x)} = f'(0) \ \Rightarrow \ \ln [f(x)] = f'(0)x + c$$

$f(x) = e^{f'(0)x} \cdot e^c$

if $\quad x = 0, f(0) = 1 \ \Rightarrow \ e^c = 1 \ \Rightarrow \ f(x) = e^{kx}$ where $k = f'(0)$

if $\quad f(0) = 0$ then put $y = 0$ in f rule $f(x) = f(x) \cdot f(0) = 0 \ \Rightarrow \ f(x) = 0.$

Q.4 Function f(x) is a differentiable function satisfying $f(x + y) = f(x) + f(y) + x^2y + y^2x \ \forall \ x, y \in R$, $f'(0) = 3$ find f(x).

Ans $f(x) = \frac{x^3}{3} + 3x$

Sol $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{f(h)+x^2h+xh^2}{h}$;

$f'(x) = 3+ x^2$; Also $f(0) = 0$; $\therefore$ $f(x) = \frac{x^3}{3} + 3x$

Q.5 Function f(x) is a differentiable function satisfying $f(xy) = f(x) + f(y)\ \forall\ x, y > 0$. $f'(1) = 3$ find f(x)

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

$f'(x) = \lim_{h\to 0} \frac{f\left(x\left(1+\frac{h}{x}\right)\right)-f(x)}{h}$

$f'(x) = \lim_{h\to 0} \frac{f(x)+f\left(1+\frac{h}{x}\right)-f(x)}{h}$

$f'(x) = \lim_{h\to 0} \frac{f\left(1+\frac{h}{x}\right)-f(1)}{x\cdot\frac{h}{x}}$ $\quad (f(1) = 0)$

$f'(x) = \frac{1}{x}\cdot f'(1) = \frac{3}{x}$

$f(x) = 3\ell n|x| + C$

$f(1) = 0 \Rightarrow c = 0$

$f(x) = 3\ell n|x|$